Chapter 1

Course Overview

- Columns, rows, matrices, $\mathbb{R}^n$, $\mathbb{R}^{m\times n}$ are reserved for representational uses.
- These represent respectively,

  vectors, adjoint vectors, linear transformations, a real vector space X of dimensions n, $\mathsf{B}(\mathsf{X},\mathsf{Y})$ between X and Y.
- As seen above, in this course, it is almost exclusively the case of a vector space over the real field $\mathbb{R}$.
  1. It is necessary to understand the geometric point of view for $\mathbb{R}^{m\times n}$, which might be obscured by using $\mathbb{C}$ field.
  2. Linear algebra over $\mathbb{R}$ is in general more complicated to study than the that over $\mathbb{C}$, and once we are done with $\mathbb{R}$ field, the corresponding material with $\mathbb{C}$ is usually immediate, but not in the other way around.
  3. We know linear algebra over finite field $\mathbb{Z}_p$ is useful in some mathematical areas. We will not cover them unfortunately.

- A vector space X can be of various kinds, that of colors, tastes, quantumn states, polynomials, $\cdots$, where the scalar multiple and addition make senses.

  For our expositional purpose, we mostly treat X as a vector space of

  $$\{\text{Locations with respect to a fixed origin}\} \quad \text{in } n\text{-dimensional physical spaces.}$$

  1. $n = 3$ case will be most intuitive for this.
  2. A location with a fixed origin can be scalar multipled, and two locations with a fixed orgin can be added by the parallelogram law.
  3. Once we attach the coordinate basis to the set of locations, a location can be specified by its coordinates, listed in column.

  $$\begin{pmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{pmatrix} \in \mathbb{R}^n.$$

Orthogonal Matrices

- We will call a matrix $O \in \mathbb{R}^{m\times n}$ orthogonal if columns of O are <u>orthonormal</u>.
- We give a definition that makes sense even when O is a rectangular matrix. If O is orthogonal, it is necessarily $m \geq n$.
- For a square orthogonal matrix $Q \in \mathbb{R}^{n\times n}$ consisting of n orthonormal columns $q_1, q_2, \cdots, q_n$, it is obvious that

  $$\begin{pmatrix} — & q_1^T & — \\ — & q_2^T & — \\ & \vdots & \\ — & q_n^T & — \end{pmatrix} \begin{pmatrix} | & | & \cdots & | \\ q_1 & q_2 & \cdots & q_n \\ | & | & \cdots & | \end{pmatrix} = \mathrm{I}.$$

  In other words, Q^T is the inverse matrix of Q.

New representation after changing basis

Let X be the n-dimensional space of locations with fixed origin.

We will refer to the current coordinate basis

$$e_1, e_2, \cdots, e_n, \quad e_i = (0, 0, \cdots, 0, 1, 0, \cdots, 0)^T \quad 1 \text{ on } i\text{-th position}$$

as *native basis*, while some other basis elements in X

$$f_1, f_2, \cdots, f_n$$

as *foreign basis*.

A location $x \in \mathsf{X}$ is currently represented by column

$$c = \begin{pmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{pmatrix}.$$

We want to know the new column whose entries represents the location x but in foreign basis

$$c' = \begin{pmatrix} x'_1 \\ x'_2 \\ \vdots \\ x'_n \end{pmatrix}.$$

To do this, (since it will be linear), we look for the native-to-foreign dictionary matrix $\square$ such that

$$c' = \square c$$

In many occasions, we are given foreign-to-native dictionary instead, that are

$$f_1, f_2, \cdots, f_n \quad \text{represented in native basis.}$$

In other words, the listing of foreign basis in columns,

$$F = \begin{pmatrix} | & | & \cdots & | \\ f_1 & f_2 & \cdots & f_n \\ | & | & \cdots & | \end{pmatrix}$$

is telling us in native language to the words written in foreign basis: $Fc' = x$ is in native language that is c.

$$Fc' = c.$$

Hence, to get c', the native-to-foregin dictionary matrix $\square = F^{-1}$.

In our numerial linear algebra course, the most important base are orthonormal ones. Thus,

$$c' = Q^T c$$

explains the procedure obtaining the new representation after changing basis that are (orthonormal) columns of Q.

Rectangular Matrices

- A matrix $A \in \mathbb{R}^{m\times n}$ represents a linear mapping A (denoted without having another symbol)
$$A : \mathsf{X} \to \mathsf{Y},$$
a member of $\mathsf{B}(\mathsf{X},\mathsf{Y})$ from X to Y of dimensions respectively n and m.

- There will be two parts of theory on rectangular matrices.
  1. $x \mapsto y = Ax$, the question is: what does A do on x to result in y?
  2. From y to the solution x. We will completely specify the set of all least square solutions.

The first part

Let A be any matrix. $A \in \mathbb{R}^{m\times n}$. To understand what does A do, we first of all wish to characterize the followings.

- The kernel of A
$$\mathsf{Ker}\,(A) = \{x \in \mathbb{R}^n \mid Ax = 0\}.$$

- The range of A
$$\mathsf{Ran}\,(A) = \{y \in \mathbb{R}^m \mid y = Ax \quad \text{for some } x \in \mathbb{R}^n\}.$$
Considering the matrix-column multiplication
$$\begin{pmatrix} | & | & | & \cdots & | \\ c_1 & c_2 & c_3 & \cdots & c_n \\ | & | & | & \cdots & | \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{pmatrix} = x_1c_1 + x_2c_2 + \cdots x_nc_n$$
we notice that
$$\mathsf{Ran}\,(A) = \mathsf{span}\,\big\langle c_1, c_2, \cdots, c_n\big\rangle.$$

- The range of A^T, the transpose of A
$$\mathsf{Ran}\,(A^T) = \{x \in \mathbb{R}^n \mid x = A^Ty \quad \text{for some } y \in \mathbb{R}^m\}.$$
If we write
$$A = \begin{pmatrix} \text{—} & r_1^T & \text{—} \\ \text{—} & r_2^T & \text{—} \\ & \vdots & \\ \text{—} & r_m^T & \text{—} \end{pmatrix} \quad \text{then} \quad \mathsf{Ran}\,(A^T) = \mathsf{span}\,\big\langle r_1, r_2, \cdots, r_m\big\rangle.$$

- The kernel of A^T
$$\mathsf{Ker}\,(A^T) = \{y \in \mathbb{R}^m \mid A^Ty = 0\}.$$

One immediately sees that

$$\begin{aligned} x \in \mathsf{Ker}\,(A) \quad &\text{if and only if} \quad x \perp \mathsf{span}\,\big\langle r_1, r_2, \cdots, r_m \big\rangle. \\ y \in \mathsf{Ker}\,(A^T) \quad &\text{if and only if} \quad y \perp \mathsf{span}\,\big\langle c_1, c_2, \cdots, c_n \big\rangle. \end{aligned}$$

In other words,

$$\mathbb{R}^n = \mathsf{Ker}\,(A) \oplus \mathsf{Ran}\,(A^T), \quad \mathbb{R}^m = \mathsf{Ker}\,(A^T) \oplus \mathsf{Ran}\,(A)$$

and they are orthogonal direct sums.

Fundamental Theorem of Linear Algebra:

Theorem 1. *For any matrix $A \in \mathbb{R}^{m\times n}$,*

$$dim\,\mathsf{span}\,\big\langle c_1, c_2, \cdots, c_n \big\rangle = dim\,\mathsf{span}\,\big\langle r_1, r_2, \cdots, r_m \big\rangle.$$

The equality of the two numbers gives rise to the definition of the rank.

- The rank of A
$$\mathsf{rank}\,(A).$$
- $\mathsf{rank}\,(A)$, $\mathsf{Ker}\,(A)$, $\mathsf{Ran}\,(A)$, $\mathsf{Ker}\,(A^T)$, and $\mathsf{Ran}\,(A^T)$ highlights about what happens in $x \mapsto Ax$.
- These will be studied in this course by
    1. QR decomposition
    2. Singular Value Decomposition.

The second part

The rootfinding part, that is to specify the complete set of all least square solutions, will be clear once we achieve QR-decomposition and Singular Value Decomposition.

Square Matrices

- In principle, all the materials for rectangular matrices applies to square matrices, where the vector spaces X and Y happen to have same dimensions.
- Inverse A^{-1} if exists, the determinant of A, $\cdots$ are additionally discussed.
- The particular part for square matrices are when we identify the domain space and the range space, that is X. $A \in \mathbb{R}^{n\times n}$ represents

$$A : \mathsf{X} \to \mathsf{X}$$

  In fancy words, $A \in \mathsf{End}(\mathsf{X})$.

  1. Considering X as space of locations, $y = Ax$ is the new location in X referenced by $x \in X$. Then we can speak of shape deformation from the referenced shape $G \subset \mathsf{X}$ by A.
  2. We can then speak of rotation, reflection, dilation, $\cdots$.
  3. Most important concept in $\mathsf{End}(\mathsf{X})$ is the invariance under A. We look for $\mathsf{M} \subset \mathsf{X}$ such that $A\mathsf{M}$ does not escape M.

Invariant vector subspace under A

- In the simplest setting, this is the eigenvalue problem.
- We will treat two cases:
  1. Diagonalization for a symmetric matrix A.
  2. Schur decomposition for any square matrix A.

Today, we examine the fact and its consequences that

- there exists a stably-working numerical algorithm implementing
- diagonalization of a real square symmetric matrix.

1. We would like to talk about why this is a nontrivial fact.
2. Believing the fact, we show we will be able to, not only define but also compute

$$\mathsf{rank}\,, \mathsf{Ran}\,, \mathsf{Ker}\,, \cdots$$

   for any $m \times n$ matrix.

Theorem. *For a $n \times n$ real symmetric matrix A, there exists an orthogonal matrix $O \in \mathbb{R}^{m\times n}$ and a diagonal matrix $\Lambda \in \mathbb{R}^{m\times n}$ such that*

$$A = O\Lambda O^T.$$

Singular Value Decomposition

- Here, we give a proof of the existence of Singular Value Decomposition for any matrix $A \in \mathbb{R}^{m \times n}$.
- The point of presenting this proof is to see if we can build an algorithm for this.

Theorem. *For any matrix $A \in \mathbb{R}^{m\times n}$, there exists a matrix $O \in \mathbb{R}^{m\times m}$, $\Sigma \in \mathbb{R}^{m\times n}$, and $Q \in \mathbb{R}^{n\times n}$ such that*

1. *O is an orthogonal matrix,*
2. *Q is an orthogonal matrix,*
3. *Σ is such that if $k = \min\{m, n\}$, the top left maximal $k \times k$ block is diagonal with diagonal entries*
$$\sigma_1 \geq \sigma_2 \geq \cdots \geq \sigma_k \geq 0$$
and all other entries are zero,
4. *$A = O\Sigma Q^T$.*

Proof. .

1. We may give a proof for the case $m \geq n$. If not, we achieve
$$A^T = O\Sigma Q^T \implies A = Q\Sigma^T O^T$$
which is in the form stated. Assume now $m \geq n$.
2. We consider a $n \times n$ symmetric matrix $A^T A$. We first check that
$$A^T Ax = 0 \quad \iff \quad Ax = 0.$$
This is because
$$A^T Ax = 0 \implies x^T A^T Ax = 0 \implies |Ax|^2 = 0 \implies Ax = 0.$$
Converse is easy.
3. $A^T A$ is a real symmetric and nonnegative definite. Hence, we have
$$A^T A = Q\Lambda Q^T.$$
The diagonalization can be arranged so that diagonal entries of Λ are in descending order:
$$\lambda_1 \geq \lambda_2 \geq \cdots \geq \lambda_n \geq 0.$$
Note that if any of $\lambda_i = 0$,
$$0 = \lambda_i q_i = A^T A q_i \implies Aq_i = 0.$$
We let $r \leq n$ be the number such that
$$\lambda_r > 0 \quad \text{and} \quad j > r \implies \lambda_j = 0.$$
and let $I_r = \{1, 2, \cdots, r\}$. The set may be empty if $r = 0$.

4. Now, define $\mathbb{R}^m$-vectors for $i \in I_r$:

$$\text{for } i \in I_r \quad o_i = \frac{Aq_i}{\sqrt{\lambda_i}}.$$

We check

$$\text{for } i, j \in I, \quad o_i^T o_j = \frac{q_i^T A^T A q_j}{\sqrt{\lambda_i \lambda_j}} = \frac{\lambda_j q_i^T q_j}{\sqrt{\lambda_i \lambda_j}} = \begin{cases} 0 & \text{if } i \neq j \\ 1 & \text{if } i = j \end{cases}$$

5. If $\{r+1, r+2, \cdots, m\}$ is nonempty, supplement o_{r+1}, o_{r+2}, $\cdots$, o_m so that

$$o_1, o_2, \cdots, o_m \quad \text{are orthonormal.}$$

6. Let O, Q, Σ be such that: O consists of columns $o_1, o_2, \cdots, o_m$, Q consists of columns $q_1, q_2, \cdots q_n$, and Σ is diagonal in the sense that entries are zero except the top left diagonals

$$\sigma_i = \sqrt{\lambda_i}, \quad i \in I_r.$$

Finally, we show that

$$Ax = O\Sigma Q^T x \quad \text{for every } x \in \mathbb{R}^n.$$

Since $q_1, q_2, \cdots, q_n$ forms a basis of $\mathbb{R}^n$, let us suppose

$$x = c_1 q_1 + c_2 q_2 + \cdots c_n q_n.$$

In other words, $c = Q^T x$. Then

$$\begin{aligned} Ax &= \sum_{i=1}^{n} c_i A q_i \\ &= \sum_{i=1}^{r} c_i A q_i && (Aq_i = 0 \text{ for } i > r) \\ &= \sum_{i=1}^{r} o_i(\sigma_i c_i) && (\text{definition of } o_i) \\ &= \sum_{i=1}^{m} o_i(\sigma_i c_i) && (\sigma_i = 0 \text{ for } i > r) \\ &= \sum_{i=1}^{m} o_i(\Sigma Q^T x)_i \\ &= O\Sigma Q^T x. \end{aligned}$$

□

For a given any matrix $A \in \mathbb{R}^{m \times n}$, we are given a Singular Value Decomposition

$$A = O\Sigma Q^T.$$

Then we can <u>compute</u>

- $\mathsf{rank}\,(A) = r$, where r is the number of nonzero singular values. If singular values are in descending order,
$$\sigma_r > 0 \quad \text{and} \quad i > r \implies \sigma_i = 0.$$
- $\mathsf{Ker}\,(A) = \mathsf{span}\left\langle q_1, q_2, \cdots q_r \right\rangle$.
- $\mathsf{Ran}\,(A) = \mathsf{span}\left\langle o_1, o_2, \cdots o_r \right\rangle$.
- $\mathsf{Ker}\,(A)^\perp = \mathsf{Ran}\,(A^T) = \mathsf{span}\left\langle q_{r+1}, q_{r+2}, \cdots q_n \right\rangle$.
- $\mathsf{Ran}\,(A)^\perp = \mathsf{Ker}\,(A^T) = \mathsf{span}\left\langle o_{r+1}, o_{r+2}, \cdots, o_m \right\rangle$.

These computations contrast their <u>definitions</u>

- $\mathsf{rank}\,(A)$, the number of independent columns (= number of independent rows)
- $\mathsf{Ker}\,(A) = \{x \in \mathbb{R}^n \mid Ax = 0\}$.
- $\mathsf{Ran}\,(A) = \{y \in \mathbb{R}^m \mid y = Ax \quad \text{for some } x \in \mathbb{R}^n\}$.
- and those for A^T.

Examine the steps 1-6 of the proof of the Singular Value Decomposition. Will we be able to build an algorithm?

It will still be a long journey until we achieve an algorithm for the diagonalization of a real symmetric square matrix.